11.14 Parabola, Not Catenary

WORKSHEET

The question

STEP I 1991, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s1-q10

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself At each joint A_i, three forces act: the tensions in the two neighbouring segments, and the weight w hanging from the strut. Which component of tension can the weight not affect?

STAGE 1 The equations at each joint

Ask yourself What happens at the middle joint A_{n+1}? Use the symmetry of the bridge.

STAGE 3 The parabola

Ask yourself Let j count joints outwards from the middle, so that A_{n+1-j} is at horizontal distance jh from A_{n+1}. How far is it above A_{n+1}?

Try this next

1. **An even number of struts.** Repeat the question with $2n$ struts, so that the lowest point of the chain is a horizontal segment in the middle. Do the joints still lie on a parabola?

2. **The tension.** Find the greatest tension in the chain, in terms of H, w and n. Where is it?

3. **The catenary.** Verify that $y = c\cosh\frac{x}{c}$ satisfies $y'' = \frac{1}{c}\sqrt{1+y'^2}$. Show that the length of the catenary from $x = 0$ to $x = X$ is $c\sinh\frac{X}{c}$.

4. **Galileo's error.** A chain of length $2L$ hangs between two supports at distance $2d$ apart. For L only slightly larger than d, show that the sag of the catenary is approximately the same as that of a parabola. Why did Galileo's experiments not reveal the difference?

Further reading

- **E. H. Lockwood, *A Book of Curves*** (Cambridge University Press). The catenary, the parabola and their relatives.
- **Jacques Heyman, *The Stone Skeleton*** (Cambridge University Press). How masonry arches and domes stand up, from Hooke's chain onwards.
- **David Billington, *The Tower and the Bridge*** (Princeton University Press). The art and

engineering of bridges.

The full guided solution, and what the question leads to at university, is in Section 11.14:
https://beyondstep.keithmcnulty.org/ch11/bridge-parabola.html